Radicals & Exponents

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

For the model $f(x) = 5x + 6$ in multiplying polynomials, evaluate $f(5)$.

Final Answer: 31. Substitute $x = 5$: $f(5) = 5(5) + 6 = 31$.

2 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

For the model $f(x) = 6x + 7$ in special products, evaluate $f(2)$.

Final Answer: 19. Substitute $x = 2$: $f(2) = 6(2) + 7 = 19$.

3 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

A data set recorded for special products consists of the four values $\{12, 16, 20, 26\}$. What is the arithmetic mean? Show your complete algebraic work.

Final Answer: 18.5. The sum is $12 + 16 + 20 + 26 = 74$, and dividing by 4 gives 18.5.

4 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

For the model $f(x) = 2x + 8$ in factoring out the greatest common factor (gcf), evaluate $f(2)$.

Final Answer: 12. Substitute $x = 2$: $f(2) = 2(2) + 8 = 12$.

5 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

A data set recorded for factoring out the greatest common factor (gcf) consists of the four values $\{13, 17, 21, 27\}$. What is the arithmetic mean? Show your complete algebraic work.

Final Answer: 19.5. The sum is $13 + 17 + 21 + 27 = 78$, and dividing by 4 gives 19.5.

6 [Foundational] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

For the model $f(x) = 3x + 9$ in factoring trinomials, evaluate $f(2)$.

Final Answer: 15. Substitute $x = 2$: $f(2) = 3(2) + 9 = 15$.

7 [Foundational] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

A data set recorded for factoring trinomials consists of the four values $\{14, 18, 22, 28\}$. What is the arithmetic mean? Show your complete algebraic work.

Final Answer: 20.5. The sum is $14 + 18 + 22 + 28 = 82$, and dividing by 4 gives 20.5.

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Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Solve $\sqrt{3x+1} = x - 1$, checking for extraneous solutions.

Square both sides: $3x + 1 = (x-1)^2 = x^2 - 2x + 1 \Rightarrow x^2 - 5x = 0 \Rightarrow x = 0$ or $x = 5$. Check $x = 0$: $1 = -1$ ✗ (extraneous). Check $x = 5$: $4 = 4$ ✓. Solution: $x = 5$.

9 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

[Radicals & Rational Exponents · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

Substitute $t = 8$: $g(8) = 5(8) + 5 = \mathbf{45}$. Setting $5t + 5 = 50$ gives $5t = 45 \Rightarrow t = \mathbf{9}$.

10 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Find the circumference of a circle with $r = 7$. Take $\pi = \frac{22}{7}$.

Apply the circumference formula $C = 2\pi r$: $C = 2 \times \frac{22}{7} \times 7$ Cancel the 7s in numerator and denominator: $C = 2 \times 22 = \mathbf{44}$ Key Mathematical Principle & Verification: In exam revision, always verify that the final value satisfies the domain constraints, units, and original governing equation by direct back-substitution into the prompt.

11 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Solve the equation $5e^{0.04t} = 45$ for t in terms of natural logarithms, and find its value rounded to two decimal places.

Step 1: Isolate the exponential expression: $e^{0.04t} = \frac{45}{5} = 9$. Step 2: Take the natural logarithm of both sides: $\ln(e^{0.04t}) = \ln(9) \Rightarrow 0.04t = \ln(9)$. Step 3: Solve for t:

$t = \frac{\ln(9)}{0.04} = 25\ln(9) = 25\ln(3^2) = 50\ln(3) \approx 50(1.0986) = \mathbf{54.93}$. The exact solution is $\mathbf{50\ln(3)}$ (approximately 54.93).

12 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

Level 3 · AdvancedA school administrator randomly surveys 120 high school seniors at Oakwood High School. In the sample, 65% report planning to attend a four-year college, with a margin of error of ±4%. Which of the following conclusions is most statistically sound? (A) Exactly 65% of all Oakwood High seniors will attend college. (B) It is plausible that between 61% and 69% of all Oakwood High seniors plan to attend a four-year college. (C) Between 61% and 69% of all high school seniors in the state plan to attend college. (D) To reduce the margin of error by half, the administrator should survey 240 seniors.

Correct Answer: (B). The margin of error creates a plausible confidence interval $65\% \pm 4\% = [61\%, 69\%]$. Because the random sample was drawn exclusively from Oakwood High seniors, the conclusion generalizes only to that school's senior class (eliminating C). To halve the margin of error, sample size must quadruple ($4 \times 120 = 480$), making D incorrect.

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13 [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Level 3 · AdvancedUsing the student prep table (Studied: 54 passed, 6 failed; Did not study: 18 passed, 22 failed; Total passed: 72 out of 100), are the events 'passed' and 'studied' independent?

Step 1: Calculate marginal probability: $P(\text{Passed}) = \frac{72}{100} = 0.72$. Step 2: Compare with conditional probability: $P(\text{Passed} \mid \text{Studied}) = \frac{54}{60} = 0.90$. Since $P(\text{Passed} \mid \text{Studied}) = 0.90 eq 0.72 = P(\text{Passed})$, the events are not independent (studying is positively associated with passing).

14 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

A cuboid has a length of USD 8$cm, a width of USD 5$ cm, and a height of USD 3 cm. Calculate the volume of the cuboid.

The volume of a cuboid is calculated using the formula: Volume = length × width × height. Given: length = USD 8$cm, width = USD 5$ cm, height = USD 3$cm. Volume = 8 \text{ cm} \times 5 \text{ cm} \times 3 \text{ cm} Volume = 40 \text{ cm}^2 \times 3 \text{ cm} Volume = 120 \text{ cm}^3 . Therefore, the volume of the cuboid is USD 120 cm^3.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 3.01 Exponents & Polynomials

A line has slope 3 and passes through $(0, 2)$. Find y when $x = 2$.

Step 1: Identify the line's equation. Since $(0, 2)$ is the y-intercept, $b = 2$. Given slope $m = 3$: $y = mx + b \Rightarrow y = 3x + 2$ Step 2: Evaluate at $x = 2$: $y = 3(2) + 2 = 6 + 2 = \mathbf{8}$ Alternative Delta Method: For every 1 unit increase in x, y increases by 3. From $x = 0$ to $x = 2$ is $+2$ units, so $y = 2 + 2(3) = \mathbf{8}$.

16 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 4.01 Functions & Transformations

Find the area of a triangle with base 10 and height 6.

Apply the triangle area formula $A = \frac{1}{2}bh$: $A = \frac{1}{2} \times 10 \times 6 = 5 \times 6 = \mathbf{30}$ square units Key Mathematical Principle & Verification: In exam revision, always verify that the final value satisfies the domain constraints, units, and original governing equation by direct back-substitution into the prompt.

17 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 5.01 Exponential & Logarithmic Functions

Evaluate the complex expression $\frac{(3-2i)^2}{1+i}$ and write the result in standard form $a + bi$.

Step 1: Expand the numerator: $(3-2i)^2 = 3^2 - 2(3)(2i) + (2i)^2 = 9 - 12i + 4(-1) = 5 - 12i$. Step 2: Multiply by conjugate $1 - i$: $\frac{5-12i}{1+i} = \frac{(5-12i)(1-i)}{(1+i)(1-i)} = \frac{5-5i-12i+12i^2}{1-i^2} = \frac{5-17i-12}{1-(-1)} = \frac{-7-17i}{2}$. In standard form: $\mathbf{-\frac{7}{2} - \frac{17}{2}i}$.

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18 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 6.01 Systems & Inequalities

In triangle XYZ, $XY = 7$ cm, $YZ = 10$ cm, and angle $XYZ = 65^\circ$. Calculate (a) the length of XZ to 3 significant figures and (b) the area of triangle XYZ.

(a) Use the Cosine Rule: $XZ^2 = 7^2 + 10^2 - 2(7)(10)\cos(65^\circ) = 49 + 100 - 140(0.4226) = 149 - 59.166 = 89.834$

$XZ = \sqrt{89.834} \approx 9.4779\ldots \approx 9.48$ cm (3 s. f.) (b) Area of triangle:

$\text{Area} = \frac{1}{2}ab\sin C = \frac{1}{2}(7)(10)\sin(65^\circ) = 35(0.9063) \approx 31.7$ cm^2 (3 s. f.)

19 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 7.01 Sequences & Series

Solve the quadratic equation $2x^2 + 5x - 4 = 0$. Give your solutions correct to 2 decimal places.

Identify coefficients: $a = 2$, $b = 5$, $c = -4$. $x = \frac{-5 \pm \sqrt{5^2 - 4(2)(-4)}}{2(2)} = \frac{-5 \pm \sqrt{25 + 32}}{4} = \frac{-5 \pm \sqrt{57}}{4}$

$x_1 = \frac{-5 + 7.5498}{4} = \frac{2.5498}{4} \approx 0.64$ $x_2 = \frac{-5 - 7.5498}{4} = \frac{-12.5498}{4} \approx -3.14$ Solutions: $x = 0.64$ or $x = -3.14$ (both to 2 d.p.).

20 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 8.01 Factoring Polynomials

Level 3 · AdvancedA quadratic function $f(x) = ax^2 + bx + c$ passes through the points $(-2, 11)$, $(0, 3)$, and $(2, 7)$. What is the value of $a + b + c$?

Solution via Desmos Table Regression:Click the + button and add a Table. Enter $x_1 = \{-2, 0, 2\}$ and $y_1 = \{11, 3, 7\}$.In the next expression line, type the regression template: y1 ~ a*x1^2 + b*x1 + c.Desmos instantly calculates the parameters: $a = 1$, $b = -1$, and $c = 3$.Calculate the target value: $a + b + c = 1 + (-1) + 3 = 3$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

For the model $f(x) = 3x + 4$ in adding and subtracting polynomials, evaluate $f(2)$.

a. 10 ✓

b. 13

c. 6

d. 14

Substitute $x = 2$: $f(2) = 3(2) + 4 = 10$.

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22 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

For the model $f(x) = 4x + 5$ in multiplying a polynomial by a monomial, evaluate $f(2)$.

a. 13 ✓

b. 17

c. 8

d. 17

Substitute $x = 2$: $f(2) = 4(2) + 5 = 13$.

23 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A data set recorded for adding and subtracting polynomials consists of the four values $\{18, 22, 26, 32\}$. What is the arithmetic mean?

a. 22.5

b. 23.5

c. 24.5 ✓

d. 26.5

The sum is $18 + 22 + 26 + 32 = 98$, and dividing by 4 gives 24.5.

24 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A data set recorded for multiplying a polynomial by a monomial consists of the four values $\{10, 14, 18, 24\}$. What is the arithmetic mean?

a. 14.5

b. 15.5

c. 16.5 ✓

d. 18.5

The sum is $10 + 14 + 18 + 24 = 66$, and dividing by 4 gives 16.5.

25 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

For a constant k, the equation $\sqrt{2x + k} = x - 1$ has $x = 5$ as a valid solution. What is the extraneous solution produced when solving this equation by squaring both sides?

a. -5

b. -1 ✓

c. 1

d. 6

We solve the problem in two main steps: 1. **Find the parameter k:** Since $x = 5$ is a solution to the original radical equation, substitute $x = 5$ into the equation: $\sqrt{2(5) + k} = 5 - 1$ $\sqrt{10 + k} = 4$ Squaring both sides gives: $10 + k = 16 \Rightarrow k = 6$ 2. **Solve the equation and find the extraneous solution:** Substitute $k = 6$ back into the equation: $\sqrt{2x + 6} = x - 1$ Square both sides to eliminate the radical: $2x + 6 = (x - 1)^2$ $2x + 6 = x^2 - 2x + 1$ Rearrange all terms to one s.

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26 [Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)
Consider the equation $\sqrt[3]{25-x^2}+\sqrt[3]{x^2+3}=4$ What is the product of all real solutions to this equation?

a. -24 ✓

b. 24

c. -48

d. 0

To solve the equation, let $u=25-x^2$ and $v=x^2+3$. Observe that the sum of the radicands eliminates x^2: $u+v=(25-x^2)+(x^2+3)=28$ The original equation can be written in terms of rational exponents: $u^{1/3}+v^{1/3}=4$ Let $a=u^{1/3}$ and $b=v^{1/3}$. Then: $a+b=4$ and $a^3+b^3=28$ Using the algebraic identity $a^3+b^3=(a+b)^3-3ab(a+b)$: $28=4^3-3ab(4)$ $28=64-12ab$ $12ab=36\Rightarrow ab=3$ Since $a+b=4$ and $ab=3$, a and.

27 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)
Consider the function $g(x)=\sqrt{x^2-6x+9}+\sqrt{x^2+8x+16}$. Find the minimum possible value of $g(x)$.

a. 2

b. 3

c. 4

d. **7** ✓

First, we simplify the terms under the square roots. The first term is $\sqrt{x^2-6x+9}$. This is a perfect square trinomial, which factors as $(x-3)^2$. Therefore, $\sqrt{x^2-6x+9}=\sqrt{(x-3)^2}=|x-3|$. The second term is $\sqrt{x^2+8x+16}$. This is also a perfect square trinomial, which factors as $(x+4)^2$. Therefore, $\sqrt{x^2+8x+16}=\sqrt{(x+4)^2}=|x+4|$. So, the function can be rewritten as $g(x)=|x-3|+|x+4|$. This expression represents the sum of the distances from x to 3 an.

28 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)
Which expression is equivalent to $(16x^8y^4)^{\frac{3}{4}}$ for all positive values of x and y?

a. $2x^2y$

b. $8x^6y^3$ ✓

c. $12x^6y^3$

d. $16x^6y^3$

To simplify the expression $(16x^8y^4)^{\frac{3}{4}}$, apply the power of a product rule $(abc)^n=a^nb^nc^n$ and the power of a power rule $(a^m)^n=a^{mn}$: 1. Evaluate the numerical coefficient: $16^{\frac{3}{4}}=\left(\sqrt[4]{16}\right)^3=2^3=8$ 2. Apply the exponent to the variable x: $(x^8)^{\frac{3}{4}}=x^{8\cdot\frac{3}{4}}=x^{\frac{24}{4}}=x^6$ 3. Apply the exponent to the variable y.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

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29 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

Consider the equation $\sqrt{2x+k} = x-1$, where k is a constant. If $x=5$ is a valid solution to the equation, what is the extraneous solution that arises when solving the equation algebraically by squaring both sides?

a. -1 ✓

b. 1

c. -5

d. 6

Step 1: Find the value of the parameter k by substituting the valid solution $x=5$ into the equation: $\sqrt{2(5)+k} = 5-1 \Rightarrow \sqrt{10+k} = 4$ Squaring both sides gives $10+k=16$, so $k=6$. Step 2: Substitute $k=6$ back into the equation and solve by squaring both sides: $\sqrt{2x+6} = x-1$ $2x+6 = (x-1)^2 = x^2-2x+1$ $x^2-4x-5=0$ Factoring the quadratic equation gives: $(x-5)(x+1)=0 \Rightarrow x=5$ or $x=-1$ Step 3: Check for extraneous sol.

30 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Let $f(x) = \sqrt{x^2-6x+9}$ and $g(x) = x-3$. For which values of x is $f(x) = g(x)$?

a. All real numbers

b. **All real numbers** $x \geq 3$ ✓

c. All real numbers $x \leq 3$

d. No real numbers

First, simplify the expression for $f(x)$. We recognize that the radicand, x^2-6x+9, is a perfect square trinomial, which can be factored as $(x-3)^2$. Therefore, $f(x) = \sqrt{(x-3)^2}$. The definition of the square root of a square is $\sqrt{a^2} = |a|$. Applying this, we get $f(x) = |x-3|$. Now, we need to solve the equation $f(x) = g(x)$, which becomes $|x-3| = x-3$. The absolute value of an expression is equal to the expression itself if and only if the expression is non-negative. Thus, $|x-3| = x-3$ if.

31 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

The graph of the function $f(x) = a\sqrt{x-h}+k$ is shown below. The function passes through the points $(5, 10)$ and $(13, 16)$. Determine the value of a and the coordinates of the vertex (h, k).

a. $a=3, (h,k)=(1,1)$

b. $a=2, (h,k)=(1,4)$

c. $a=3, (h,k)=(1,4)$ ✓

d. $a=2, (h,k)=(1,1)$

The vertex of the graph is (h, k). From the graph, we can infer that the vertex is at $(1, 4)$. Thus, $h=1$ and $k=4$. The function can be written as $f(x) = a\sqrt{x-1}+4$. We are given that the graph passes through the point $(5, 10)$. Substituting these values into the function: $10 = a\sqrt{5-1}+4$. This simplifies to $10 = a\sqrt{4}+4$, so $10 = 2a+4$. Subtracting 4 from both sides gives $6=2a$, which means $a=3$. Therefore, $a=3$ and the vertex is $(h,k)=(1,4)$.

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32 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

What is the value of the expression $\sqrt{36} + \sqrt{49}$?

a. 10

b. 11

c. **13** ✓

d. 15

First, calculate the square root of 36. Since $6^2 = 36$, $\sqrt{36} = 6$. Next, calculate the square root of 49. Since $7^2 = 49$, $\sqrt{49} = 7$. Finally, add the two results: $6 + 7 = 13$.

33 [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Check: is $x = -1$ a solution of $\sqrt{x} = -1$?

a. Maybe

b. Only in $\mathbb{C}$

c. **No (square root is non-negative)** ✓

d. Yes

Applying Radicals & Rational Exponents principles, $\sqrt{x} \geq 0$ for real $x \geq 0$. Therefore, the verified correct option is No (square root is non-negative).

34 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Write $x^{3/2}$ as a radical.

a. $x^{1/2} \cdot x$

b. $x\sqrt{x^2}$

c. $\sqrt{x^3}$ **or** $(\sqrt{x})^3$ ✓

d. $\sqrt[3]{x^2}$

Applying Radicals & Rational Exponents principles, $x^{3/2} = (x^{1/2})^3 = \sqrt{x^3}$. Therefore, the verified correct option is $\sqrt{x^3}$ or $(\sqrt{x})^3$.

35 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

Find $f(x) = \sqrt{x} + \sqrt{1 - x}$. What is the domain?

a. $x \geq 0$

b. All reals

c. $0 < x < 1$

d. $0 \leq x \leq 1$ ✓

Applying Radicals & Rational Exponents principles, Need both $x \geq 0$ and $1 - x \geq 0$. Therefore, the verified correct option is $0 \leq x \leq 1$.

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36 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Write $\sqrt[3]{a^2b}$ using rational exponents.

a. $a^{2/3}b^{1/3}$ ✓

b. $a^{1/2}b^{1/3}$

c. $a^{3/2}b^{1/3}$

d. a^2b^3

Applying Radicals & Rational Exponents principles, $(a^2b)^{1/3} = a^{2/3}b^{1/3}$. Therefore, the verified correct option is $a^{2/3}b^{1/3}$.

37 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Which of the following equals $8^{-2/3}$?

a. $\frac{1}{8}$

b. 4

c. $\frac{1}{4}$ ✓

d. -4

Applying Radicals & Rational Exponents principles, $8^{-2/3} = 1/(8^{2/3}) = 1/4$. Therefore, the verified correct option is $\frac{1}{4}$.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 6.01 Systems & Inequalities

Find the real number(s) equal to $(-32)^{3/5}$.

a. 8

b. Undefined

c. -8 ✓

d. -32^3

Applying Radicals & Rational Exponents principles, $(-32)^{1/5} = -2$. $(-2)^3 = -8$. Therefore, the verified correct option is -8.

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39 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 7.01 Sequences & Series

Write $\sqrt[4]{x^3}$ using rational exponent.

a. x^{3+4}

b. $4x^3$

c. $x^{4/3}$

d. $x^{3/4}$ ✓

Applying Radicals & Rational Exponents principles, $\sqrt[n]{x^m} = x^{m/n}$. Therefore, the verified correct option is $x^{3/4}$.

40 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 8.01 Factoring Polynomials

Simplify $\sqrt{4}$ — note: solve $\sqrt{36}$.

a. 9

b. 6 ✓

c. 3

d. 12

Applying Radicals & Rational Exponents principles, $\sqrt{4 \cdot 9} = 2 \cdot 3 = 6$. Therefore, the verified correct option is 6.

41 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 1.01 Linear Equations & Lines

Determine the number of real solutions of $\sqrt{x} = x^2 - 6x + 9$.

a. 0

b. 1

c. 4

d. 2 ✓

$\sqrt{x} = (x-3)^2$. Let $u = \sqrt{x}$: $u = (u^2 - 3)^2$. Solutions exist at $x = 0$ and $x = 4$ (verify graphically).

42 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 10.01 Rational Expressions

Solve: $\sqrt{2x+3} = x$. Check for extraneous roots.

a. $x = 3$ **only** ✓

b. $x = -1$ only

c. $x = 3$ and $x = -1$

d. No solution

$2x + 3 = x^2 \Rightarrow x^2 - 2x - 3 = (x-3)(x+1) = 0$. $x = -1$: $\sqrt{1} = 1eq - 1$. Extraneous. $x = 3$: $\sqrt{9} = 3$. ✓.

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43 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 11.01 Statistics & Data Analysis

Prove $\sqrt[n]{ab} = \sqrt[n]{a} \cdot \sqrt[n]{b}$ using rational exponents.

a. True only for $n = 2$

b. $(ab)^{1/n} = a^{1/n}b^{1/n}$ **— follows from power rules for real numbers ✓**

c. Not always true

d. Requires $a, b > 0$

Real number power rules: $(ab)^{1/n} = a^{1/n}b^{1/n}$ for $a, b \geq 0$.

44 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 2.01 Quadratic Functions

Simplify: $\frac{\sqrt{x+h} - \sqrt{x}}{h}$ (rationalize numerator).

a. $\frac{1}{2\sqrt{x}}$

b. $\frac{1}{\sqrt{x+h} + \sqrt{x}}$ ✓

c. $\frac{h}{\sqrt{x+h} + \sqrt{x}}$

d. $\frac{\sqrt{x+h} - \sqrt{x}}{h}$

Multiply by $\frac{\sqrt{x+h} + \sqrt{x}}{\sqrt{x+h} + \sqrt{x}}$.

45 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 3.01 Exponents & Polynomials

Express $\sqrt[6]{x^4}$ in simplest radical form.

a. $\sqrt[3]{x^2}$

b. $\sqrt[6]{x^4}$

c. $x^{2/3}$

d. Both A and C ✓

$x^{4/6} = x^{2/3} = \sqrt[3]{x^2}$. A and C are equal.

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